

Timothy Do

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EDUCATION

University of California, Los Angeles

Sep. 2023 – Jun. 2024

M.S. Electrical and Computer Engineering, Signal Processing Specialization

Los Angeles, CA

- **Cumulative GPA:** 3.933/4.0
- **Relevant Coursework:** Computational Imaging, Digital Speech Processing, Advanced Deep Learning & Neural Networks, Modern Wireless Communication Systems, Large-Scale Social Complex Networks
- **Clubs and Societies:** ECEGAPS (Electrical and Computer Engineering Graduate and PostDoc Society)

University of California, Irvine

Sep. 2019 – Jun. 2023

B.S. Electrical Engineering, Digital Signal Processing & Communications Specialization

Irvine, CA

- **Cumulative GPA:** 3.924/4.0, **Major GPA:** 3.944/4.0, **Magna Cum Laude, Accelerated Status**
- **Relevant Coursework:** Computer Architecture, Control Systems, Digital Image/Signal Processing, Electronics
- **Clubs and Societies:** Irvine Computer Vision Laboratory, IEEE, IPC, Eta Kappa Nu, Tau Beta Pi

EXPERIENCE

Senior Developer Evangelist

Jan. 2026 - Present

SiMa.ai

San Jose, CA

- Led 4+ interns for deploying Physical AI applications. Brought up the deployment of VLAs (ACT, SmolVLA, GR00T-N1.6) on the Modalix platform and integrating it with lerobot for subsecond inference.
- Developed SiMa.ai's GenAI experience: Palette Neat GenAI Studio. Supports 30+ LLMs/VLMs from Hugging Face, Whisper transcription, and Supersonic 3 TTS running concurrently on the Modalix platform.
- Created a 10+ video series to support existing users on the Palette Neat platform. Each 5-15 min module covers topics like SDK setup, going through model compilation, and application examples, etc.
- Facilitated customer facing workshops in different sites across the U.S., Japan, South Korea, Taiwan and Singapore. Showcased SiMa.ai's Palette Neat Agentic SDK to hundreds of prospective customers.

Developer Evangelist

Oct. 2025 - Jan. 2026

SiMa.ai

San Jose, CA

- Created SiMa.ai's first Developer Community, which grew to 500+ users in a year. Managed posts, announced software releases, and created the public support channel for existing users with over 50+ threads answered.
- Maintained SiMa.ai's developer documentation, from carrier board bringup for Modalix to setup the Palette Neat framework to develop applications. Developed auto-translation framework on ollama for docs in 5+ languages.
- Generated visibility by posting technical content on behalf of the company on LinkedIn, generating 5000+ impressions with 100+ likes per post marketing SiMa.ai's platform.

Computer Vision Software Engineer

Nov. 2024 - Jan. 2026

SiMa.ai

San Jose, CA

- Developed and showcased demos like scratch defect detection on cars and label verification via OCR at high traction conferences such as DeveloperWeek, Embedded Vision, TSMC Symposium, etc.
- Led GStreamer-based projects for OCR, Change Detection, and Visual Language Models, managing the full pipeline from data collection and model finetuning to quantization and deployment.
- Supported customers end to end, from the optical selection, imaging process, to perform factory acceptance testing and validation through 50+ product lines running on SiMa.ai MLSoC.

Computer Vision Intern

Aug. 2024 - Nov. 2024

SiMa.ai

San Jose, CA

- Deployed and presented a production-level Anomaly Detection demo on SiMa.ai's MLSoC at the 2024 AI Hardware & Edge AI Summit within two weeks of joining.
- Led project tasks including data collection, model finetuning, and deployment for Anomaly and Change Detection, while pitching technical solutions to industry leaders. Project yielded >\$1M in revenue.
- Pitched to C-suite on SiMa.ai's value proposition from a technical perspective in being the premier solution in providing a chip down solution through their MLSoC platform to empower end to end edge AI pipelines.

Embedded Software Engineering Co-Op

Jun. 2023 – Apr. 2024

Maxeon Solar Technologies

San Jose, CA

- Implemented Robust OpenCV pipelines to quantitatively detect module features for defect binning
- Developed Tkinter GUI interfaces for operator ease of use when testing images under custom data pipelines
- Trained robust YOLOv5/YOLOv8 Module Defect Classification Models deployed in manufacturing fab

Lead Undergraduate Student Researcher

Jan. 2022 – Jun. 2023

Irvine Computer Vision Laboratory at UC Irvine

Irvine, CA

- Researched Spectral Image Filtering and Baseball Simulations under Professor Glenn Healey
- Digitized Hyperspectral Optical Imaging Filters into CSVs and compared them against certain chemical emissions
- Managed 15+ Graduate/Undergraduate Researchers in Google Spaces and developed shared MATLAB/Python Scripts hosted on Google Drive/Github for analyzing SWIR/NVIR vegetation spectras

Industrial Internet of Things Intern

Jun. 2022 – Sep. 2022

Western Digital

San Jose, CA

- Applied Industrial Toolsets to develop a Realtime Raspberry Pi Sensor Data Analytics Platform
- Earned Six Sigma Yellow Belt award for learning foundations of Lean Six Sigma through LinkedIn Learning
- Saved the company an annual projected cost of \$312,000 from hazard mitigations in the manufacturing fab

PROJECTS

End-to-End Automatic Modulation Recognition | *Python, MATLAB, Pytorch*

Jan. 2024 – Mar. 2024

- Designed a CNN-based AMR System on I/Q samples (93.2% test accuracy at high SNR)
- Preprocessed multiple modulation datasets by one-hot encoding integer labels for minimizing cross-entropy loss
- Applied data augmentation by index reversal on each of the data samples for a 3% performance improvement

African-American Dialect Classification | *Python, Librosa, Spafe, Opensmile*

Feb. 2024 – Mar. 2024

- Classified different dialects of African Americans across the east coast with the COORAL dataset
- Developed a XGBoost Classifier using Opensmile and GFCC features for 95% test accuracy
- Applied chatter noise augmentation to the samples resulting in improving accuracy by 2%
- Implemented multiprocessing pipeline to extract multiple samples simultaneously (3-4X speedup)

EEG Action Classification | *Python, Tensorflow, Pytorch*

Feb. 2024 – Mar. 2024

- Explored various model architectures on EEGs to classify a subject's actions, the best yielding 75.4% test accuracy
- Performed Data Cleaning on the BCI dataset by trimming and maxpooling to improve classification accuracy
- Optimized model hyperparameters by automating exhaustive grid search on learning rate, batch size, and epochs

SKILLS

Electronics: Raspberry Pi, Network Analysis, Cadence, LTSpice, Soldering, Oscilloscope & Multimeter Measurements

Programming Languages: Python, Java, C, CUDA, MATLAB, bash, MySQL HTML/CSS/Javascript, VHDL

Developer Tools: git, SSH, Makefile, Linux, Azure, VS Code, Visual Studio, PyCharm, Eclipse, Vivado HLS

Libraries: OpenCV, Pandas, Numpy, Matplotlib, PIL, Tensorflow, Pytorch, Tkinter, Socket, Flask, Librosa